

Agentic AI & Multiagent Systems



Semester Course of 03 Credits

B.Tech. (CSE, DSAI) 6th Sem / M.Tech. (CSE)
2nd Sem

Prof Giovanni Merlino

Associate Professor (Computer Engineering)

University Of Messina, Italy

Prof Dr. Om Prakash Vyas

Vice Chancellor & Director

IIIT Naya Raipur-CG, India

Outline

- Course Delivery: IIITNR & Uni of Messina, Italy
 - **Prof Om Prakash Vyas & Team from IIITNR**
 - **Prof Giovanni Merlino and Teaching Assistants from UniME, Italy.**
- Study material & Reference Books
- Agentic AI Overview: What, Why and How of AI Agents
 - Agentic AI Vs Traditional Programs
 - Agentic AI Vs Objects
- Evolution of AI Agents: Chronological developments
- Drivers of Agentic AI
 - Large Language Model, Advanced Reinforcement Learning & Multimodal Integration
- Agentic AI Projects: Overview

Course Delivery & Resources

The Course on Intelligent agents and multi-agent systems is being organised jointly with Univ of Messina, Italy (Deptt of Computer Engineering)

- Lectures in hybrid mode alongwith weekly assignments and periodical Quiz
- Project work on AI Agents by 'Group of Students' towards research directions.
- Study material of sessions to be collected compiled and distributed by Dr Kavita Jaiswal in coordination with UniME Team
- Collaborative Learning Mode: Students Group also studies and Presents
- Online sessions to be coordinated by UniME Team
- Course delivery and Evaluation to be done jointly
- Compilation & Submission of Attendance and Grades by IITNR



- **IIITs** in India are very specialized Universities set up as **Center of excellence** by Government of India, started with **Indian Institute of Information Technology, Allahabad**.
- Currently there are **25 IIITs** working in IT *Super-specializations* in **AI, Machine Learning, Data Science, Business Informatics, IoT, Robotics/Drone** and **Interdisciplinary** areas to be explored further...
- In the Central India, newly established **IIIT** is functional since 2015 in Smart city **Naya Raipur....**



Brief about Prof Giovanni Merlino & Team

(Deptt of Computer Engineering, Uni of Messina, Italy)

- Prof. Giovanni Merlino is an **Associate Professor** in the Department of Engineering at the **University of Messina, Italy**, with expertise in **distributed systems, IoT, edge/fog computing, and intelligent cyber-physical systems**.



- With many multi-national projects completed, he has published extensively in leading international journals and conferences and has been actively involved in European research projects. Prof. Merlino's work on **distributed systems, IoT infrastructures, edge/fog computing, and cyber-physical systems** directly relates to foundational aspects of **Agentic AI and multi-agent coordination**, particularly in **smart cities, sensor networks, and adaptive autonomous systems**.

- With their strong expertise in scalable distributed architectures and real-world experimental testbeds, Prof. Merlino and his team bring deep practical insights into agent interaction and collaborative intelligence. Their involvement will expose IIIT–Naya Raipur students to global perspectives, real-world system design challenges, and cutting-edge research directions in Agentic AI and Multi-Agent Systems.

Brief about Prof Om Prakash Vyas & Team

(IIIT Naya Raipur, India)



Dr. Kavita Jaiswal

Assistant Professor
Deptt. of CSE, IIIT NR
IoT, Cloud Computing, Fog Computing
and WSN



Prof. O. P. Vyas

Data Science, Explainable Artificial Intelligence (XAI) and Cyber Security.



Mr. Abhijeet Kar

Research Scholar (CSE)
IIIT Naya Raipur
Cyber Security

Course Objectives

- Introduce the theoretical foundations of intelligent agents and multi-agent systems
 - Evolution of AI Agents & Transformative Powers of AI Agents
 - AI Agents Taxonomy
 - Autonomy, reactivity, proactiveness, social ability
 - Agentic AI in modern systems
 - AI Agents Architectures
- Significant Projects on AI Agents:
 - OpenAI's Operator Project & STAR Project
- Study communication, coordination, cooperation, and negotiation among agents
- Explain formal models of agency, rationality, and autonomy
- Bridge classical multi-agent theory with modern agentic AI systems
- Enable students to design and implement autonomous and multi-agent systems
- Analyze ethical, safety, and alignment issues in agentic AI

Course Delivery: AI Agents-What & Why to Multi Agent Systems

A Semester course on **Agentic AI & multiagent system** represents a paradigm shift in computational thinking, where systems comprise multiple autonomous entities working independently yet interconnectedly, so the Course will have 03 major streams ;



What & Why AI Agents

The course will begin with a formal introduction to **AI agents**, including their **definition, taxonomy, and fundamental characteristics**, followed by a discussion on the need and motivation for Agentic AI. Selected **significant Agentic AI projects and real-world implementations** will then be examined to contextualize the concepts.



How does AI Agents work & Agent Architecture

It delves into the practical realities of building these intelligent systems, providing a comprehensive blueprint through the “**AI Agent Architecture**.” comparative analysis of leading agent frameworks like AutoGen, LangGraph, LlamaIndex, and AutoGPT. This goes beyond surface-level descriptions, dissecting their strengths, weaknesses, and design philosophies, particularly in areas like state management, tool integration, and decision-making logic



Foundations of Multi Agent System

Agents communicate, coordinate, and influence one another's outcomes through complex and often strategic exchanges. Accordingly, a comprehensive understanding of multi-agent systems requires mastery of four interconnected theoretical domains: **Algorithmic foundations**, **Game-theoretic foundations**, **Logical foundations**, and **Learning foundations** are essential foundations. Together, these domains provide the essential tools for analysing and designing robust, scalable, and intelligent agent interactions.

AI Agents: Definition

An agent is a computer system that is situated in some environment, and that is capable of autonomous action in this environment in order to meet its design objectives.



AI Agent: Agents vs Traditional Programs, Agents Vs Objects

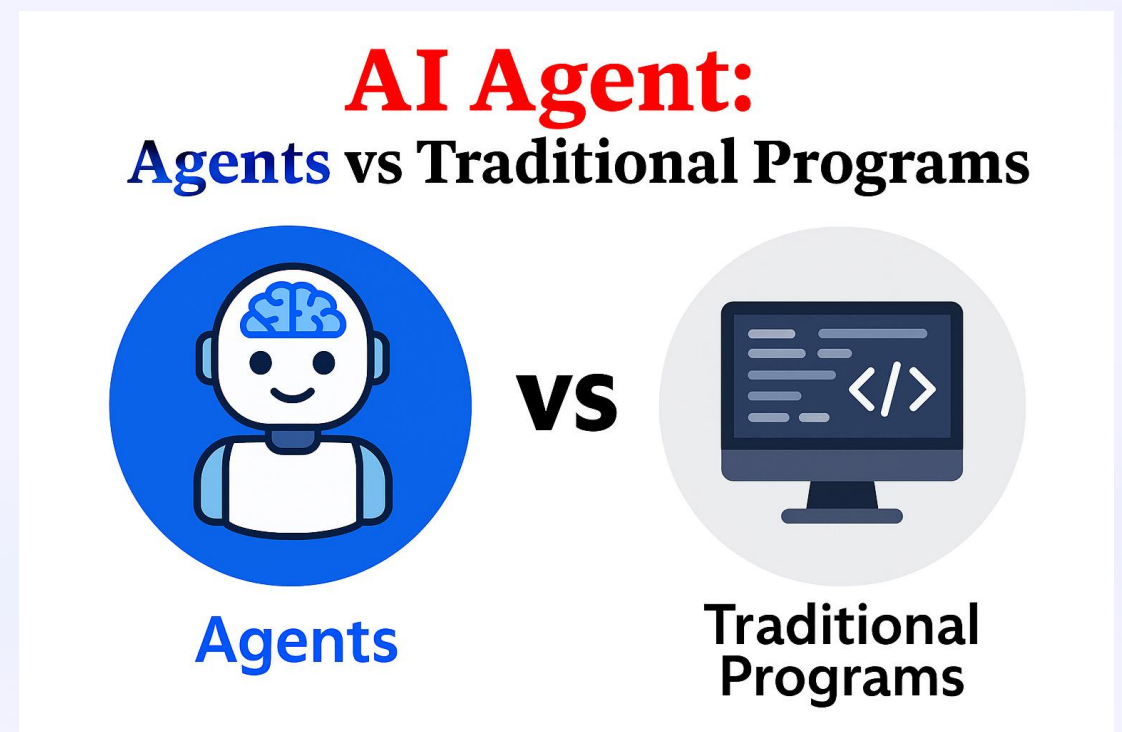
Traditional programs are sequences of instructions that execute when invoked. They are typically **deterministic and reactive**—waiting for inputs or stimuli and then executing predefined steps. Such programs **cannot act autonomously**; they require explicit invocation and follow fixed control flows.

Agents, on the other hand, are designed to **operate autonomously toward high-level goals**, continuously monitor their environment, and adjust their behavior without constant external control.

Key point:

> *Programs* execute fixed instructions in response to external triggers.

> *Agents* **decide when and how to act** based on their goals and perceptions.



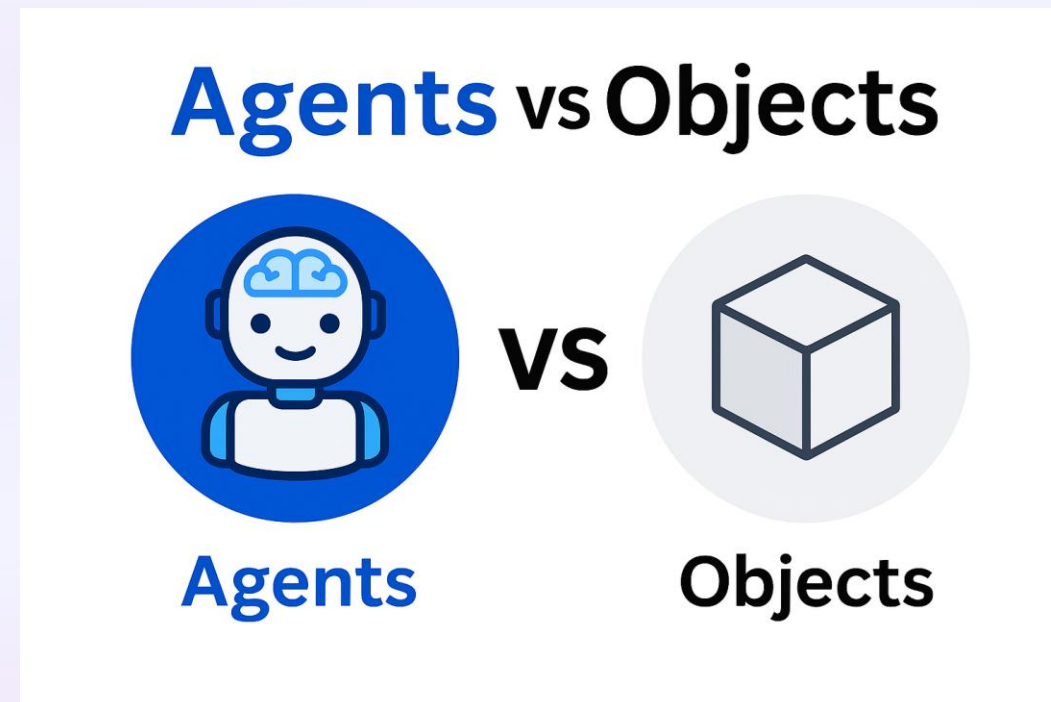
AI Agent: Agents Vs Objects

Although both are software constructs, **agents** and **objects** differ in **autonomy, behavior flexibility, and execution control**:

Autonomy

Objects are passive and perform operations only when another component explicitly invokes their methods (they have no control over their own execution).

Agents have **strong autonomy**—they decide *if* and *when* to perform actions, including refusing requests from others.



AI Agent: Agents Vs Objects

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Behavior Flexibility

Objects encapsulate state and methods but lack inherent mechanisms for flexible behavior (e.g., proactive planning or environmental adaptation).

Agents support **reactive, proactive, and social behavior**, meaning they can react to environmental changes, take initiative toward goals, and interact with other agents.

Activeness

Objects are invoked under an external control thread (e.g., the main program or calling object).

Agents are **active entities** with their own control thread or execution loop, making them capable of concurrently carrying out tasks. This is why multi-agent systems are inherently **multi-threaded** in concept.

A classic way Wooldridge (and others) summarize this is with the motto:

👉 ***“Objects do it for free; agents do it because they want to.”***

Defining the AI Agent: Beyond Passive Computation

An agent is a computer system that is capable of independent action on behalf of its user or owner. In other words, an agent can figure out for itself what it needs to do in order to satisfy its design objectives, rather than having to be told explicitly what to do at any given moment.

The quintessential AI agent transcends traditional command-response paradigms, functioning as a proactive collaborator in knowledge discovery and problem-solving. It represents a synthetic cognoscente capable of **Core Capabilities** autonomous reasoning and strategic action.

- Conducting autonomous epistemic forays across the digital noosphere
- Employing sophisticated heuristics to dissect multitiered conundrums
- Orchestrating protracted strategies for intricate objectives
- Engaging in perpetual self-optimisation through experiential learning

This quantum leap in artificial cognition heralds a seismic shift in machine intelligence applications across myriad domains of human endeavour, fundamentally altering how we conceptualise human-machine collaboration.



^SAgentic AI: AI Agents and Multi Agent Systems

Agents communicate, coordinate, and influence one another's outcomes through complex and often strategic exchanges.

A **multiagent system** is one that consists of a number of agents, which interact with one another, typically by exchanging messages through some computer network infrastructure.

In the most general case, the **agents in a multiagent system** will be representing or acting on behalf of users or owners with very different goals and motivations. In order to successfully interact, these agents will thus require the ability to cooperate, coordinate, and negotiate with each other, in much the same way that we cooperate, coordinate, and negotiate with other people in our everyday lives

Multi Agent Systems

Exploring the foundational principles of autonomous agents working together in complex environments

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Unlike conventional software systems, agents function as **independent decision-makers**, frequently operating with incomplete information about their environment and fellow agents.



What Is a Multiagent System?

A **multiagent system (MAS)** represents a paradigm shift in computational thinking, where systems comprise multiple autonomous entities working independently yet interconnectedly.



Multiple Autonomous Entities

Diverse agents operating with independence and decision-making capability



Individual Objectives

Each agent pursues distinct goals whilst navigating shared environments



Strategic Interaction

Agents communicate and influence one another's outcomes through complex exchanges

Understanding multiagent systems requires mastery across **four** interconnected theoretical domains, each providing essential tools for analysing and designing agent interactions.

HISTORICAL

CONTEXT

The Historical Trajectory of AI

Agents

Early Foundations (1950s-1970s)

Symbolic AI and rule-based systems emerged, establishing foundational concepts of autonomous computation and decision-making frameworks.

1

2

Expert Systems Era (1980s)

Knowledge-based agents demonstrated domain-specific reasoning capabilities, marking the first practical applications of agent-based paradigms.

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Reactive Paradigms (1990s)

Behaviour-based robotics and subsumption architectures challenged traditional approaches, introducing reactive and situated intelligence models.

4

Learning Agents (2000s-2010s)

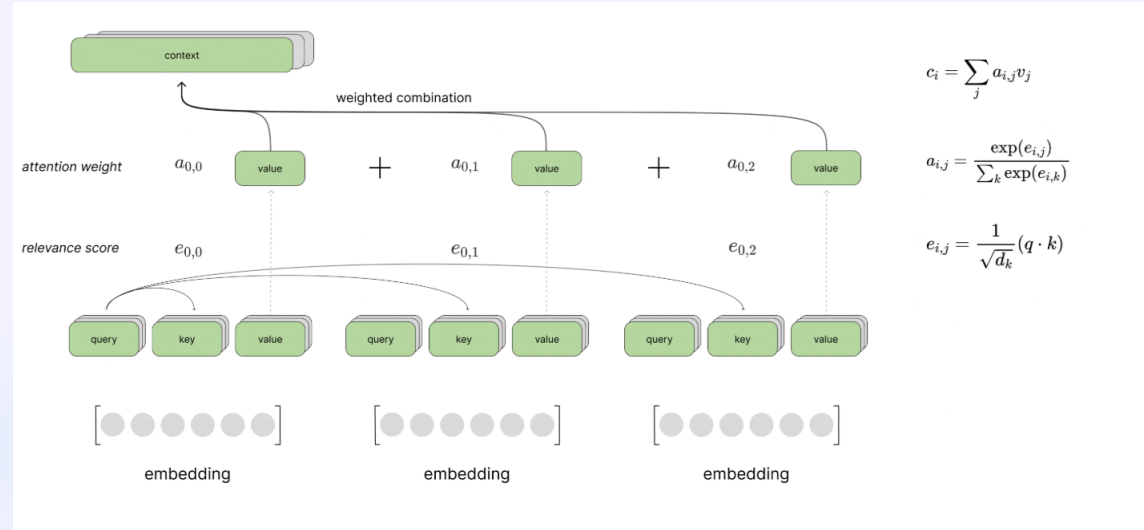
Machine learning integration enabled adaptive behaviours, with reinforcement learning facilitating autonomous skill acquisition and optimisation.

5

Contemporary Renaissance (2020s-Present)

Large language models and transformer architectures have catalysed unprecedented agentic capabilities, enabling sophisticated reasoning and multimodal interaction.

Technological Drivers of the AI Agent Renaissance



Foundational Enablers

The contemporary resurgence of agentic AI stems from convergent technological advancements across multiple research fronts, fundamentally altering the landscape of autonomous systems.



Large Language Models

Transformer-based architectures enable sophisticated natural language understanding, reasoning over textual knowledge, and few-shot task adaptation capabilities previously unattainable.



Advanced Reinforcement Learning

Deep RL algorithms facilitate autonomous skill acquisition in complex environments, enabling agents to learn optimal policies through environmental interaction and reward optimisation.



Multimodal Integration

Vision-language models and cross-modal architectures allow agents to perceive, reason about, and act upon information across diverse sensory modalities simultaneously.



Technological Drivers: Infrastructure and Tooling



Computational Infrastructure

Distributed computing platforms and specialised hardware accelerators provide the computational substrate necessary for training and deploying sophisticated agent architectures at scale.



Memory Architectures

External memory systems, vector databases, and retrieval-augmented generation enable agents to access vast knowledge repositories dynamically during inference.



Tool Integration Frameworks

Standardised APIs and function-calling mechanisms allow agents to interact with external systems, databases, and computational resources programmatically.

Transformative potential of AI agents

Sophisticated entities capable of autonomous action, complex reasoning, and adaptive learning. This evolution, driven by unprecedented computational power, advances in NLP, big data, and algorithmic breakthroughs, marks a shift from reactive AI to systems exhibiting near- human cognitive abilities.

Key Insights

Paradigm Shift:

AI agents represent a move from tool-based AI to collaborative partners, poised to revolutionize industries and redefine human-machine interaction.

Confluence of Technologies:

The current AI agent renaissance is fuelled by a unique convergence of technological advancements, creating a fertile ground for unprecedented innovation.

From Narrow to General:

The development trajectory is toward increasingly general-purpose agents capable of cross-domain learning and problem-solving, exemplified by projects like **OpenAI's OperatorAgent** and **Stanford's STaR**.

Self-Improvement as a Cornerstone: STaR's self-teaching capabilities highlight a crucial trend toward AI systems that can autonomously learn and adapt, minimizing the need for human intervention.

Societal Transformation, not just technological:

The rise of AI agents will have profound implications beyond technology, impacting work, creativity, and problem-solving and necessitating careful ethical considerations regarding their development and deployment. They are more than just tools, but collaborators